(a) Prefix reuse

Saved interaction · Qwen3-4B

Task: diagnose a timeout in client.acquire()

Reused · Restored · Computed

1 Build

Submit the original timeout prompt

0 / 534

reused / prompt tokens

2 Append

Add a regression-test request

512 / 656

reused / prompt tokens

3 Edit early

Modify an early token in the original prompt

0 / 534

reused / prompt tokens

4 Restore

Clear GPU cache; replay original prompt

512 / 534

restored / prompt tokens

Restore result — TTFT 47.3 ms

client.acquire() does not enforce the timeout.

512 restored tokens · DRAM → GPU

KV Inspector — After Build

Independent block export/import

18.00 MiB → 12.73 MiB

0 BF16 mismatches · 576 records

(b) GPU capacity

Recorded · S = 12 · NVMe 2 TiB

54 GiB KV / GPU

	Raw	Pool mode
KV tokens ↑	1.18M	1.59M
Waiting requests (mean) ↓	1.5	0.4
Engine TTFT (s) ↓	5.16	1.94

(c) History retention

Recorded · S = 8 · NVMe 640 GiB

400 GiB DRAM

	Raw	Tier mode
Cached tokens (GPU + ext.) ↑	55%	91%
Stored / raw bytes ↓	100%	71.6%
Engine TTFT (s) ↓	6.76	2.34